

LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 3

Course Code: CSE252

Course Title: INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Course Outcome: CO3

Paper: Sample Subjective Questions

Maximum Marks: 10 x 10 = 100

Unit: 3

Unit Title: Supervised Machine Learning

Unit Mapped RBT: L3 (Apply)

Total Questions: 10

Suggested Duration: Not timed - practice set

Subjective Questions (10 x 10 = 100 Marks)

Q1. Apply linear regression to a dataset and describe how the line is fitted and how it is read. (10 Marks)

- (a) State what quantity the fitting procedure minimises and what the slope and intercept represent.
- (b) Use a fitted model to predict a value, and explain what the residuals would tell you about the fit.

[L3 - Apply]

Q2. Compare a single decision tree with a Random Forest built from many such trees. (10 Marks)

- (a) State how each of the two arrives at a prediction.
- (b) Use the ideas of bias and variance to explain why the forest usually performs better on unseen data.

[L3 - Apply]

Q3. Examine a regression whose residual plot shows a clear curved pattern rather than random scatter. (10 Marks)

- (a) State what the pattern tells you about the model that was fitted.
- (b) Use two possible remedies to describe how you would proceed, and state the risk each one carries.

[L3 - Apply]

Q4. Use the ideas of bias and variance to explain the effect of raising the complexity of a model. (10 Marks)

- (a) State what each of the two contributes to the total error.
- (b) Use a named model whose complexity can be tuned to show how the two readings move as it is raised.

[L3 - Apply]

Q5. Apply the Naive Bayes classifier to the task of sorting messages into spam and not spam. (10 Marks)

- (a) State what the classifier computes for a new message and how it chooses a class.
- (b) Use the independence assumption to explain why the method still works despite being false here.

[L3 - Apply]

Q6. Compare linear regression with polynomial regression. (10 Marks)

- (a) State what each of the two can represent about the relationship in the data.
- (b) Use a small dataset to explain what happens as the degree of the polynomial is raised too far.

[L3 - Apply]

Q7. Examine a classifier that predicts the majority class for every record and still records a high accuracy. (10 Marks)

- (a) State what about the dataset makes this possible.
- (b) Use two alternative measures to describe how the failure would be exposed.

[L3 - Apply]

Q8. Use the split criterion of a decision tree to explain how the tree is constructed. (10 Marks)

- (a) State what is computed at each node in order to select a feature and a threshold.
- (b) Use this to explain why the procedure is described as greedy, and what that costs.

[L3 - Apply]

Q9. Use the distinction between a parameter and a hyperparameter to plan the fitting of a model. (10 Marks)

- (a) State which quantities are learned from the data and which must be chosen beforehand, with examples.
- (b) Use a validation partition to describe how the hyperparameters would be chosen.

[L3 - Apply]

Q10. Compare the decision boundary formed by a decision tree with the one formed by K-Nearest Neighbour. (10 Marks)

- (a) State the shape of boundary each of the two produces and why it takes that shape.
- (b) Use a dataset whose classes are separated by a diagonal line to say which of the two struggles, and why.

[L3 - Apply]

----- End of Sample Question Set -----